CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Tinkercad 4: Design a skate park piece

Goal: I can design a skate park piece that fits the size rule and has a flat bottom.

35 minutes | 7 steps | Computer Science and Programming, Digital Literacy | Playbook A meeting 27

Ask | Plan | Build | Test | Improve

You are done when

Tell a partner which piece you made and its size.

You will use

- Cart laptops, charged
- Tinkercad Classroom class code
- One printed test ramp if a printer is ready
- Test boards or finger boards if approved, cardboard boards otherwise
- Rulers and size rule card: 80 by 60 by 40 millimeters

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Sign in — 4 min

- Get your laptop.
- Open Tinkercad with the class code.

2 Step 2: Measure a board — 1 min

- Hold a test board. How wide is it?

3 Step 3: Watch the size rule — 4 min

- No piece bigger than 80 by 60 by 40 millimeters.
- Flat bottom, at least 40 millimeters wide.

4 Step 4: Design your piece — 15 min

- Choose a kicker ramp, quarter pipe, grind box or rail.
- Start from a wedge or a box.
- Check your size with the numbers.

5 Step 5: Teacher check — 4 min

- Show the flat bottom and the size.
- Fix anything flagged.

6 Step 6: Save and shut — 4 min

- Check that your work saved.
- Close the lid with two hands.
- Put the laptop back in its slot and plug it in.

7 Step 7: Show your result — 3 min

- Tell a partner which piece you made and its size.

What to turn in

A skate park piece that passes the size and flat bottom check.

No computer? Another way to do it

Draw the piece on millimeter grid paper at 1:1 and build it in thin cardboard.

Finished early? Next challenge

Add a grip texture or a small logo without breaking the size rule.

Stay safe

Nicknames or initials only. Test boards stay on tables.

Grade targets

Grade 5: Pick one piece and meet the size rule.
Grade 6: Add a grip texture or logo and still meet the size rule.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.